

$$\begin{array}{ccc}
[X, Y] & \xrightarrow{\quad = \quad} & [X, Y] \\
\downarrow \eta_{[X,Y]} & & \downarrow [X, \eta_Y] \\
& (ii) & [X, U^T F^T Y] \\
U^T F^T [X, Y] & \xrightarrow[U^T \psi_{X,Y}]{} U^T [F^T X, F^T Y]_T \xrightarrow{\quad = \quad} & [U^T F^T X, U^T F^T Y] \\
& & \uparrow [\eta_X, U^T F^T Y]
\end{array}$$